信息学院软件工程系

《计算机网络》实验报告

题　　目 实验四　CISCO IOS 路由器基本配置

班　　级 数字媒体技术 2023 级数媒班

姓　　名 马鑫

学　　号 37220232203780

实验时间 2025 年 10 月 24 日

2025 年 2 月 15 日

填写说明

1、本文件为 Word 模板文件，建议使用 Microsoft Word 2024 打开，在可填写的区域中如实填写；

2、填表时勿改变字体字号，保持排版工整，打印为 PDF 文件提交；

3、文件总大小尽量控制在 1MB 以下，最大勿超过 5MB；

4、在实验课结束 14 天内，按实验报告提交到我校课程网站的指定位置，源代码等主要材料上传在公开的代码托管平台上。

5、鼓励同学之间探讨，鼓励合理使用人工智能平台，提升效率，但不应滥用相关资源，如抄袭代码和代写作业。

1 实验目的

通过完成实验，理解网络层和路由的基本原理。掌握路由器配置网络和组网的方法；掌握 IP 协议、IP 地址配置和路由的概念；掌握 IP 协议和路由的基本原理；了解在模拟器下根据教程配置网络的方法。

2 实验环境

Windows11

3 实验结果

1. 按照附件一描述使用 Router eSIM v1.1 模拟器来模拟路由器的配置环境。

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with END.
Router(config)#hostname lab_A
```

enable 进入超级用户模式，config t 进入全局配置模式，hostname lab_A 将路由器出厂默认名字 Router 改为 lab_A。

```
lab_A(config)#banner motd #
Enter TEXT message.  End with the character '#'.
Accounting Department
#
```

修改当日的消息标题。

```
lab_A(config)#ip host lab_A 192.5.5.1 205.7.5.1 201.100.11.1
lab_A(config)#ip host lab_B 219.17.100.1 199.6.13.1 201.100.11.2
lab_A(config)#ip host lab_C 223.8.151.1 204.204.7.1 199.6.13.2
lab_A(config)#ip host lab_D 210.93.105.1 204.204.7.2
lab_A(config)#ip host lab_E 210.93
% Incomplete command.
lab_A(config)#ip host lab_E 210.93.105.2
```

Lab_A	Not Completed
Hostname	Done
Enable Secret	Not Done
Line Console Login	Not Done
Line Console Password	Not Done
Line vty Login	Not Done
Line vty Password	Not Done
E0 IP	Not Done
E0 Shutdown	Not Done
E1 IP	Not Done
E1 Shutdown	Not Done
S0 IP	Not Done
S0 Clock Rate	Not Done
S0 Shutdown	Not Done
Routing Protocol	Not Done
Network 1	Not Done
Network 2	Not Done
Network 3	Not Done
IP Host Lab_A	Done
IP Host Lab_B	Done
IP Host Lab_C	Done
IP Host Lab_D	Done
IP Host Lab_E	Done
Time elapsed	12:29

配置 ip 地址和机器名映射表。

```
lab_A(config)#int eth 0
lab_A(config-if)#ip address 192.5.5.1 255.255.255.0
lab_A(config-if)#int eth 1
lab_A(config-if)#ip address 205.7.5.1 255.255.255.0
lab_A(config-if)#int serial 0
lab_A(config-if)#ip address 201.100.11.1 255.255.255.0
lab_A(config-if)#exit
```

配置路由器接口对应的 ip 地址。

```
lab_A(config)#interface serial 0
lab_A(config-if)#clock rate 56000
lab_A(config-if)#exit
lab_A(config)#exit
00:17:16: %SYS-5-CONFIG_I: Configured from console by console
```

为串行 DCE 接口配置时钟周期。

```
lab_A#configure term
Enter configuration commands, one per line.  End with END.
lab_A(config)#interface serial 0
lab_A(config-if)#no shutdown
lab_A(config-if)#exit
lab_A(config)#exit
00:23:25: %SYS-5-CONFIG_I: Configured from console by console
```

设置端口为激活配置

```
lab_A#config t
Enter configuration commands, one per line.  End with END.
lab_A(config)#enable secret class
lab_A(config)#interface serial 0
lab_A(config-if)#no shutdown
lab_A(config-if)#shutdown
lab_A(config-if)#no shutdown
lab_A(config-if)#exit
lab_A(config)#interface ethernet 1
lab_A(config-if)#shutdown
lab_A(config-if)#no shutdown
lab_A(config-if)#exit
lab_A(config)#interface ethernet 0
lab_A(config-if)#no shutdown
lab_A(config-if)#exit
```

Lab_A	Not Completed
Hostname	Done
Enable Secret	Done
Line Console Login	Not Done
Line Console Password	Not Done
Line vty Login	Not Done
Line vty Password	Not Done
E0 IP	Done
E0 Shutdown	Done
E1 IP	Done
E1 Shutdown	Done
S0 IP	Done
S0 Clock Rate	Done
S0 Shutdown	Done
Routing Protocol	Not Done
Network 1	Not Done
Network 2	Not Done
Network 3	Not Done
IP Host Lab_A	Done
IP Host Lab_B	Done
IP Host Lab_C	Done
IP Host Lab_D	Done
IP Host Lab_E	Done
Time elapsed	28:52

完成打开关闭端口的任务。

```
lab_A(config)#interface ethernet 0
lab_A(config-if)#description engineering LAN,Bldg,1#
lab_A(config-if)#exit
lab_A(config)#line console 0
lab_A(config-line)#login
lab_A(config-line)#password cisco
lab_A(config-line)#exit
lab_A(config)#line vty 0 4
lab_A(config-line)#login
lab_A(config-line)#password software
lab_A(config-line)#password cisco
lab_A(config-line)#exit
```

为路由器设置密码。

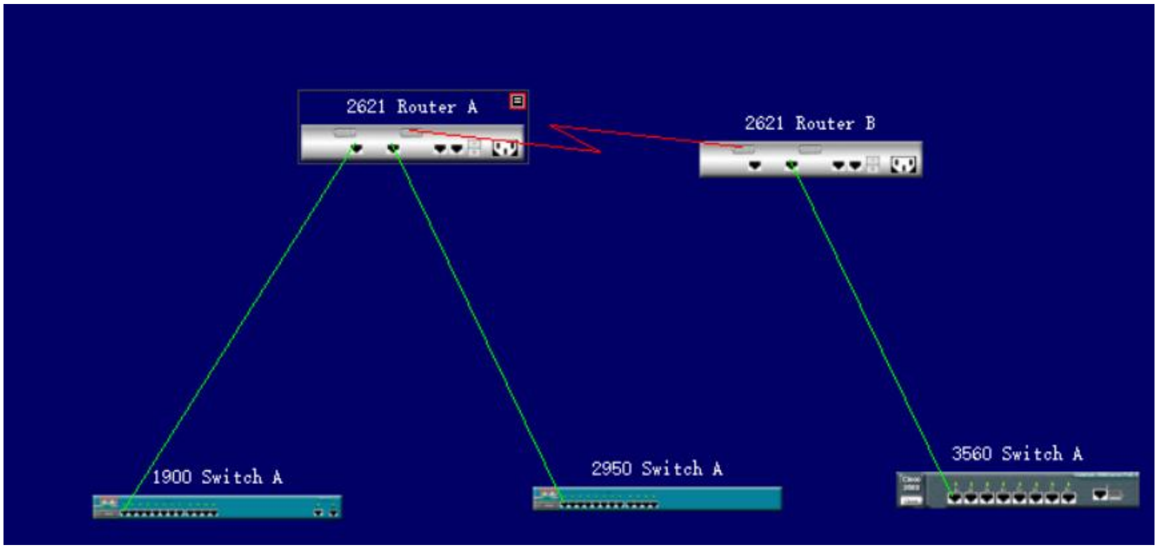
```
lab_A(config)#router rip
lab_A(config-router)#network 192.5.5.1
lab_A(config-router)#network 205.7.5.1
lab_A(config-router)#network 201.100.11.1
```

配置动态路由。

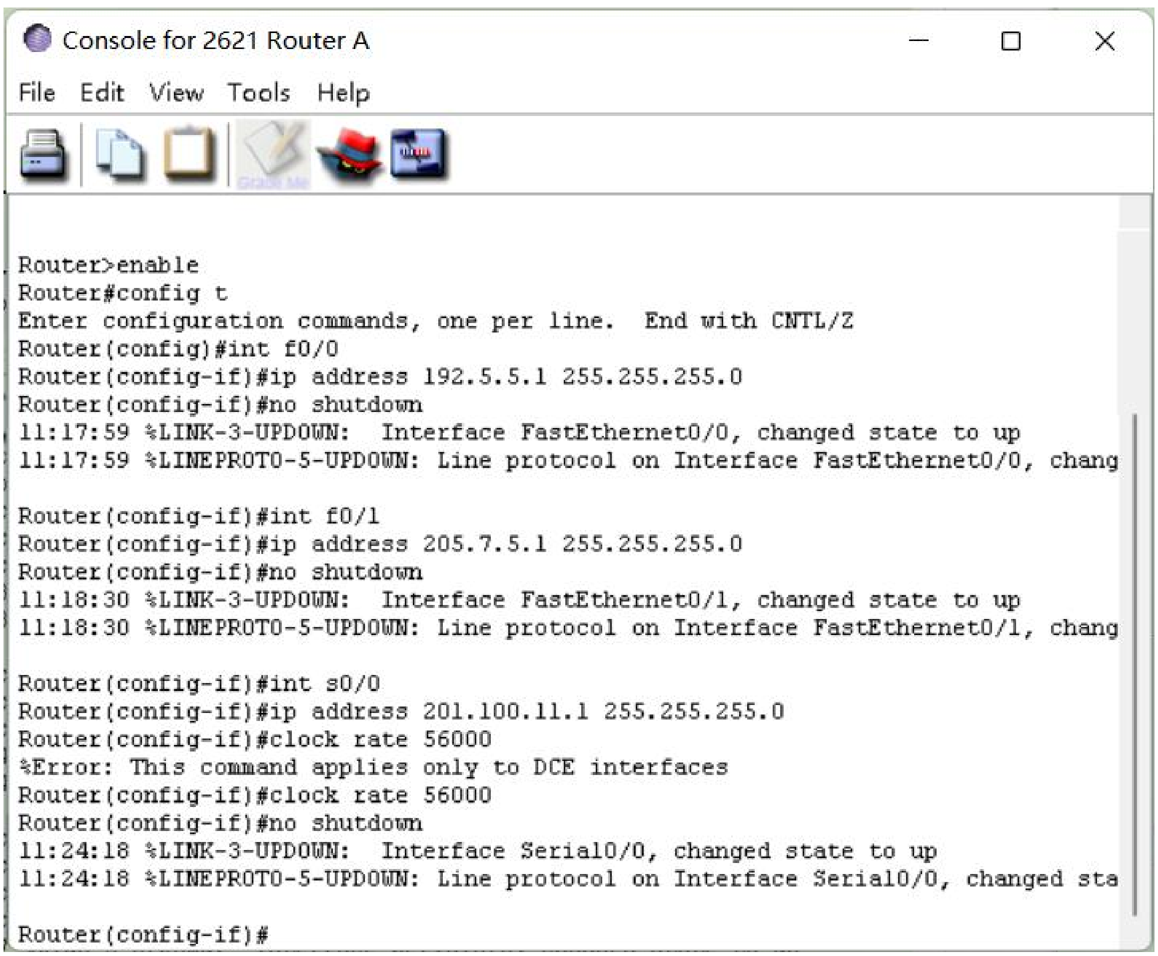
Lab_A	Completed
Hostname	Done
Enable Secret	Done
Line Console Login	Done
Line Console Password	Done
Line vty Login	Done
Line vty Password	Done
E0 IP	Done
E0 Shutdown	Done
E1 IP	Done
E1 Shutdown	Done
S0 IP	Done
S0 Clock Rate	Done
S0 Shutdown	Done
Routing Protocol	Done
Network 1	Done
Network 2	Done
Network 3	Done
IP Host Lab_A	Done
IP Host Lab_B	Done
IP Host Lab_C	Done
IP Host Lab_D	Done
IP Host Lab_E	Done
Time elapsed	38:18

完成配置。

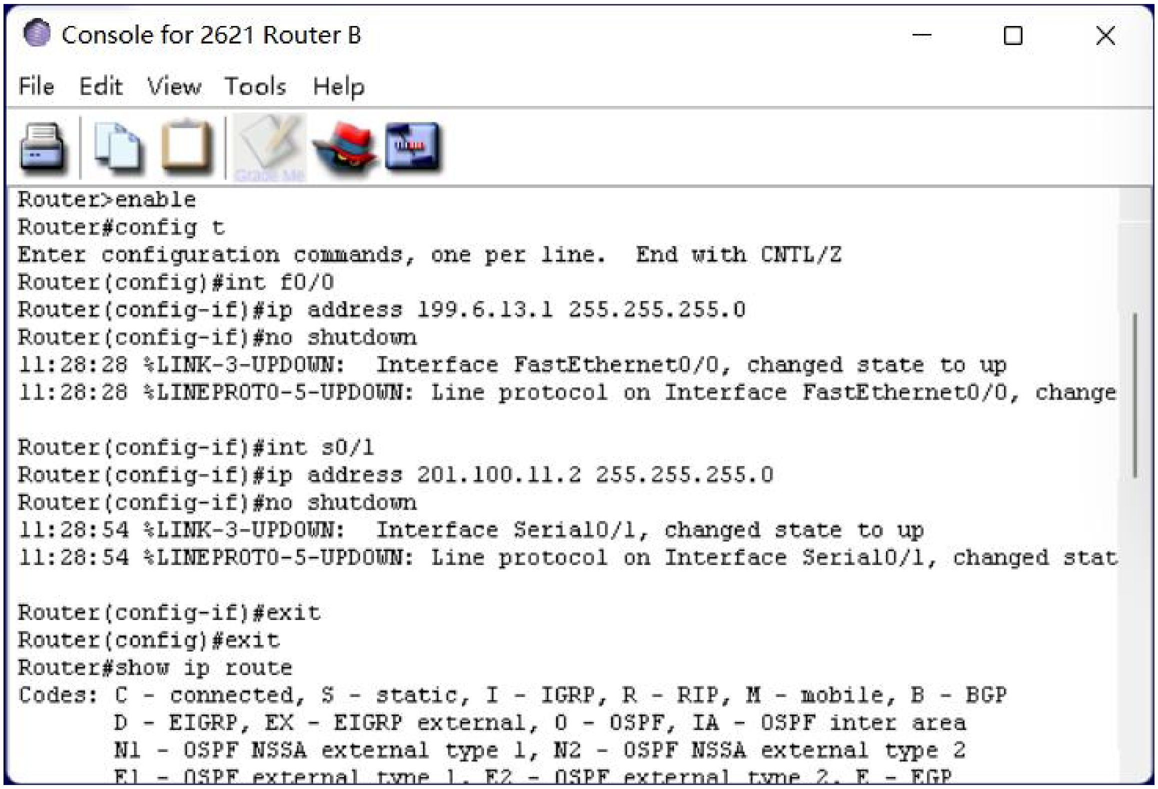
2.使用 CCNA Network Visualizer 6.0 配置静态路由、动态路由和交换机端口的VLAN（虚拟局域网）。



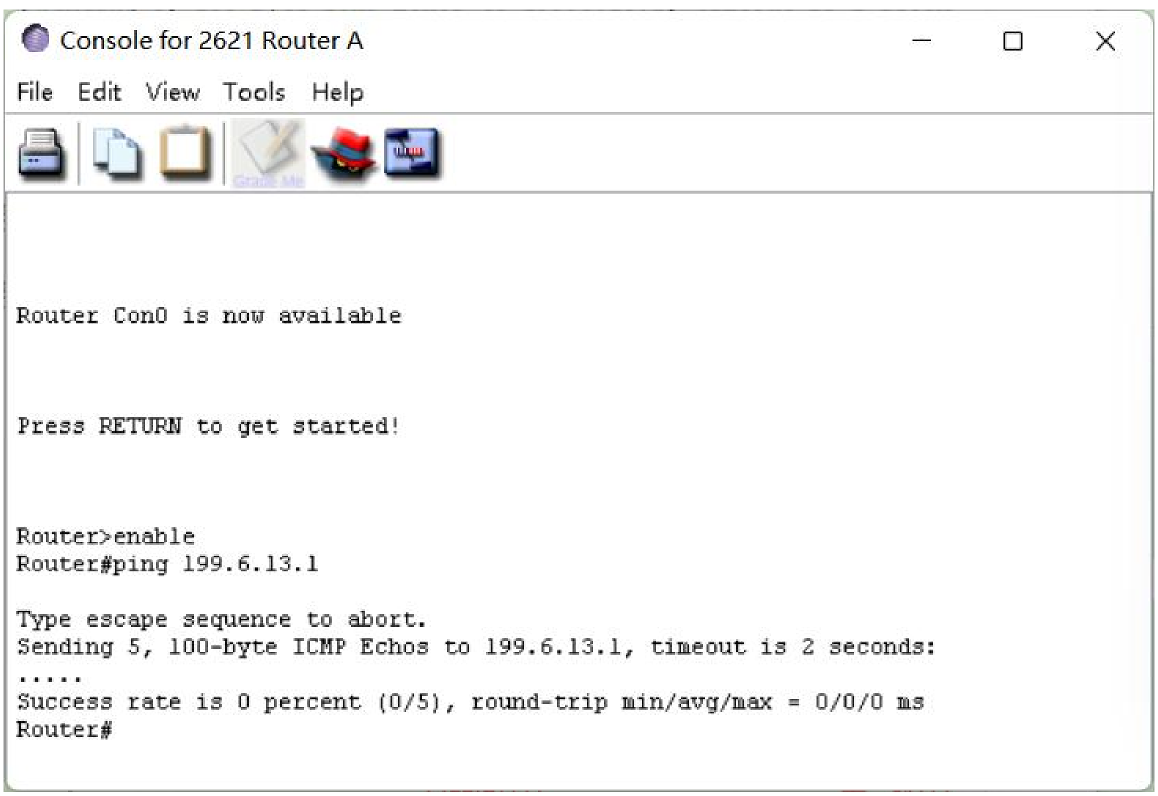
连接路由器与交换机设备。



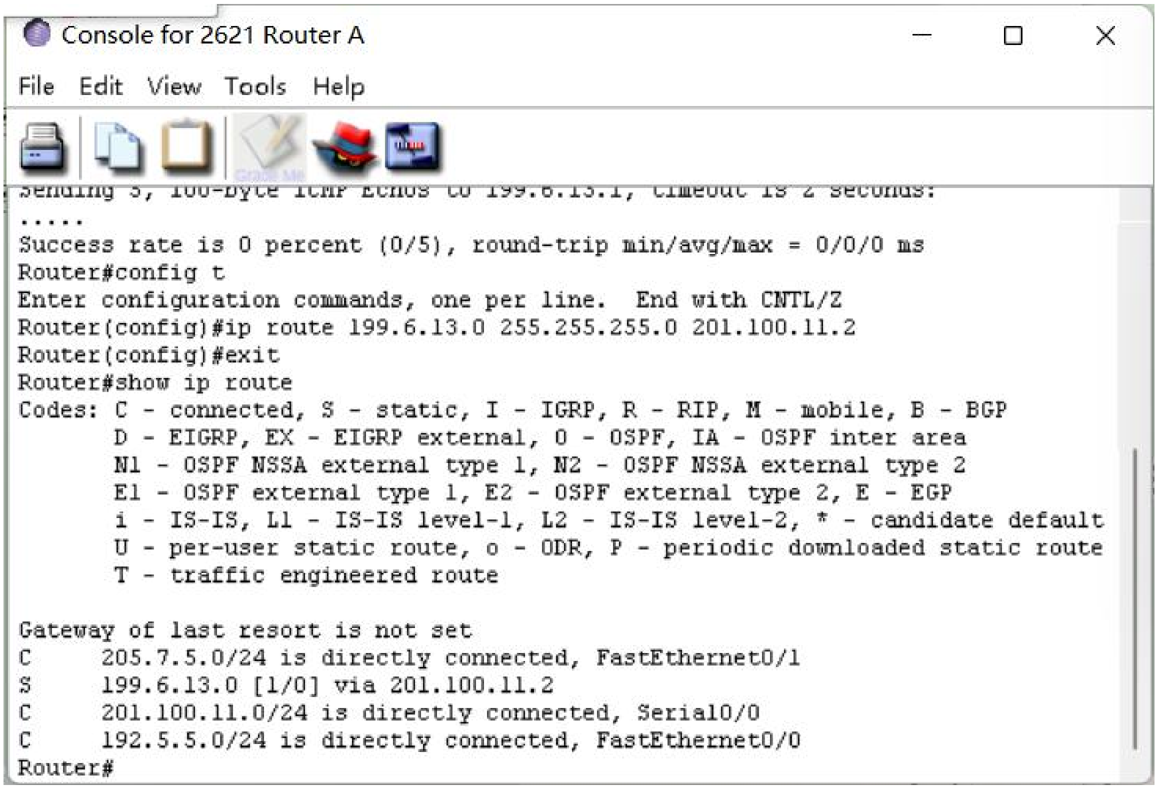
配置路由器 A 的各个端口的 IP 地址，S0/0 为 DCE 端口。



配置路由器 B 各个端口的 IP 地址，s0/1 端口为 DTE 端口。



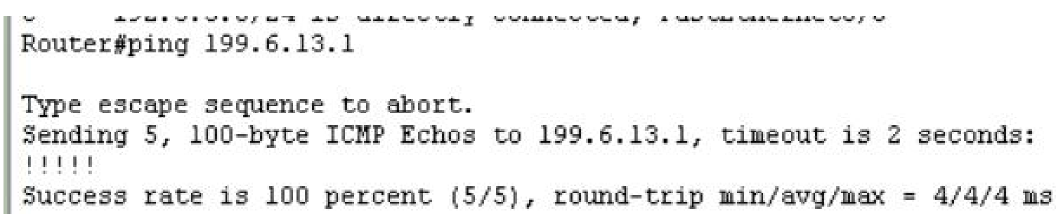
在路由器 A 使用 ping 命令查看路由器 A 到 3560 交换机 A 的直连网络是否连通，显示没有连通。




```
Sending 5, 100-byte ICMP Echos to 199.6.13.1, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5), round-trip min/avg/max = 0/0/0 ms
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z
Router(config)#ip route 199.6.13.0 255.255.255.0 201.100.11.2
Router(config)#exit
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, * - candidate default
       U - per-user static route, o - ODR, P - periodic downloaded static route
       T - traffic engineered route

Gateway of last resort is not set
C     205.7.5.0/24 is directly connected, FastEthernet0/1
S     199.6.13.0 [1/0] via 201.100.11.2
C     201.100.11.0/24 is directly connected, Serial0/0
C     192.5.5.0/24 is directly connected, FastEthernet0/0
Router#
```

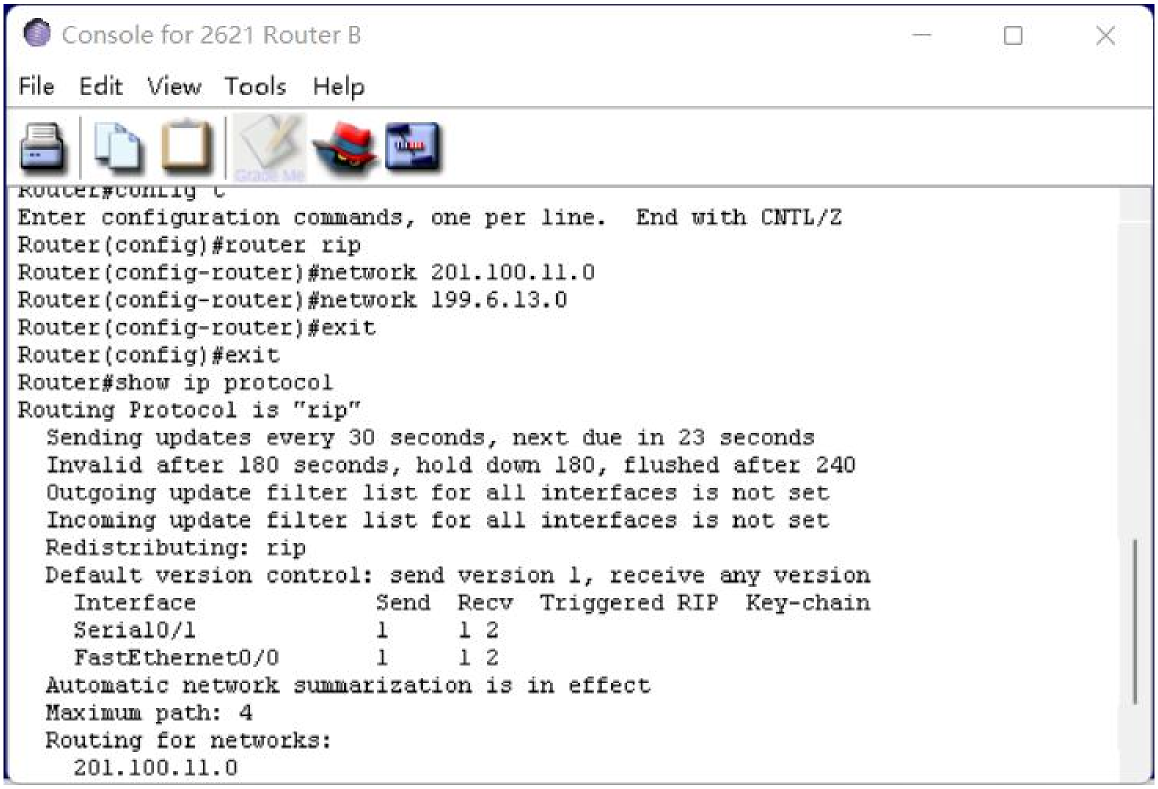
为路由器 A 配置静态路由协议。



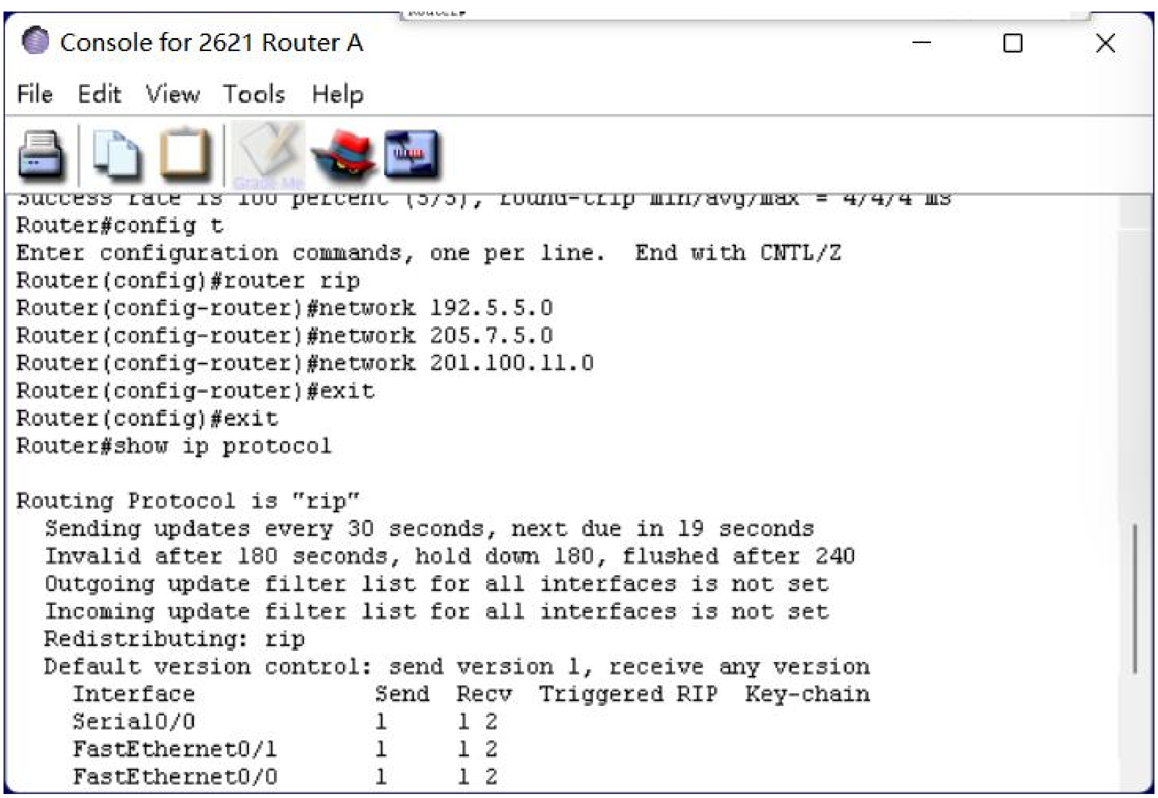
```
Router#ping 199.6.13.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 199.6.13.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/4/4 ms
```

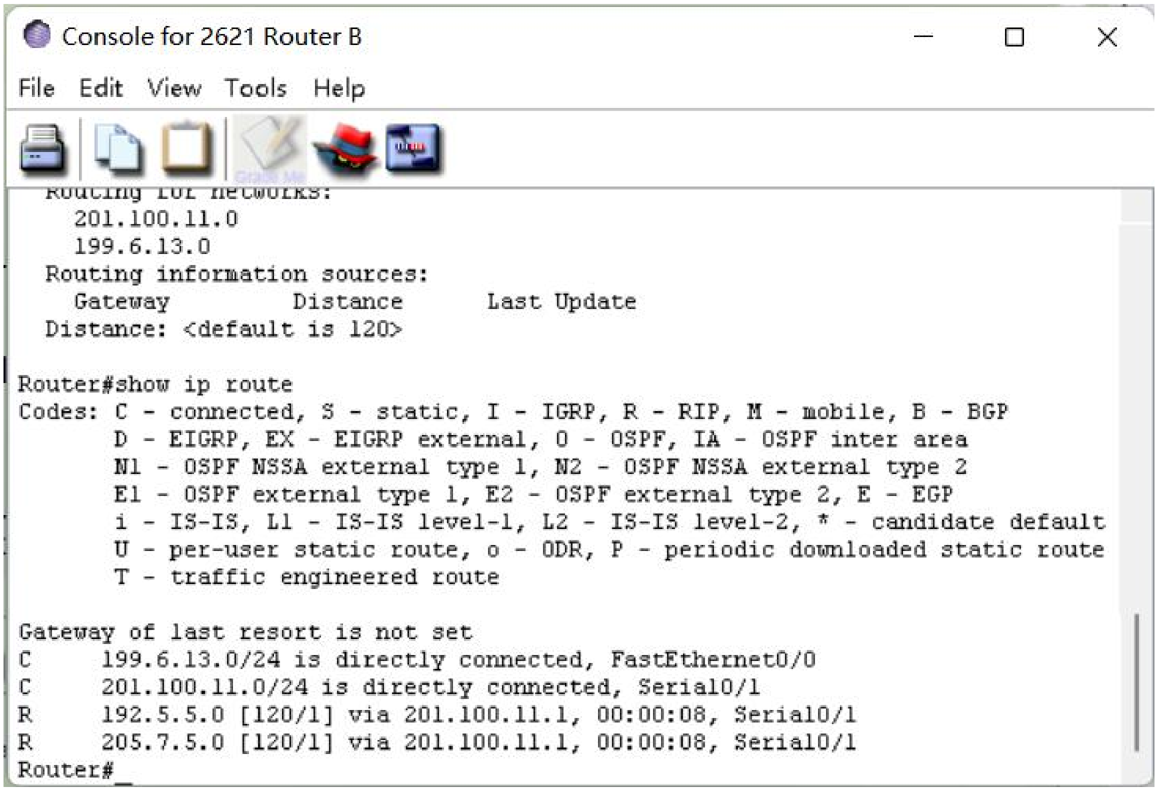
重新使用 ping 命令，连通。



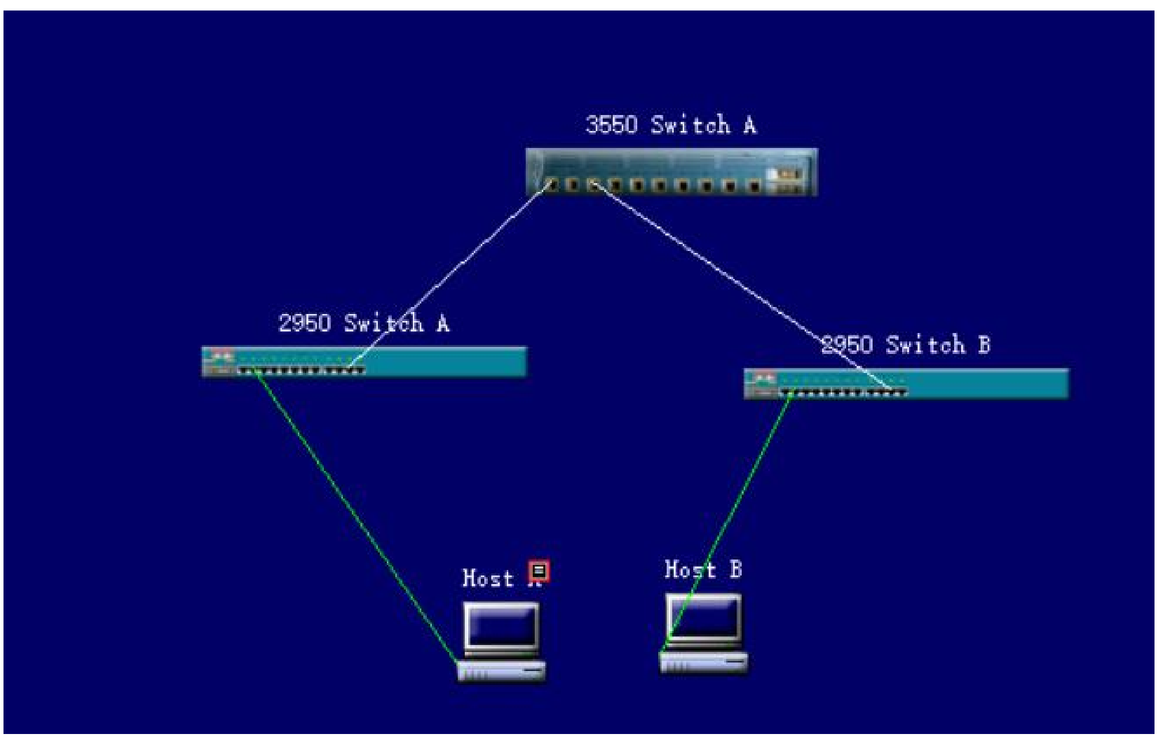
为路由器 B 配置 RIP 协议。



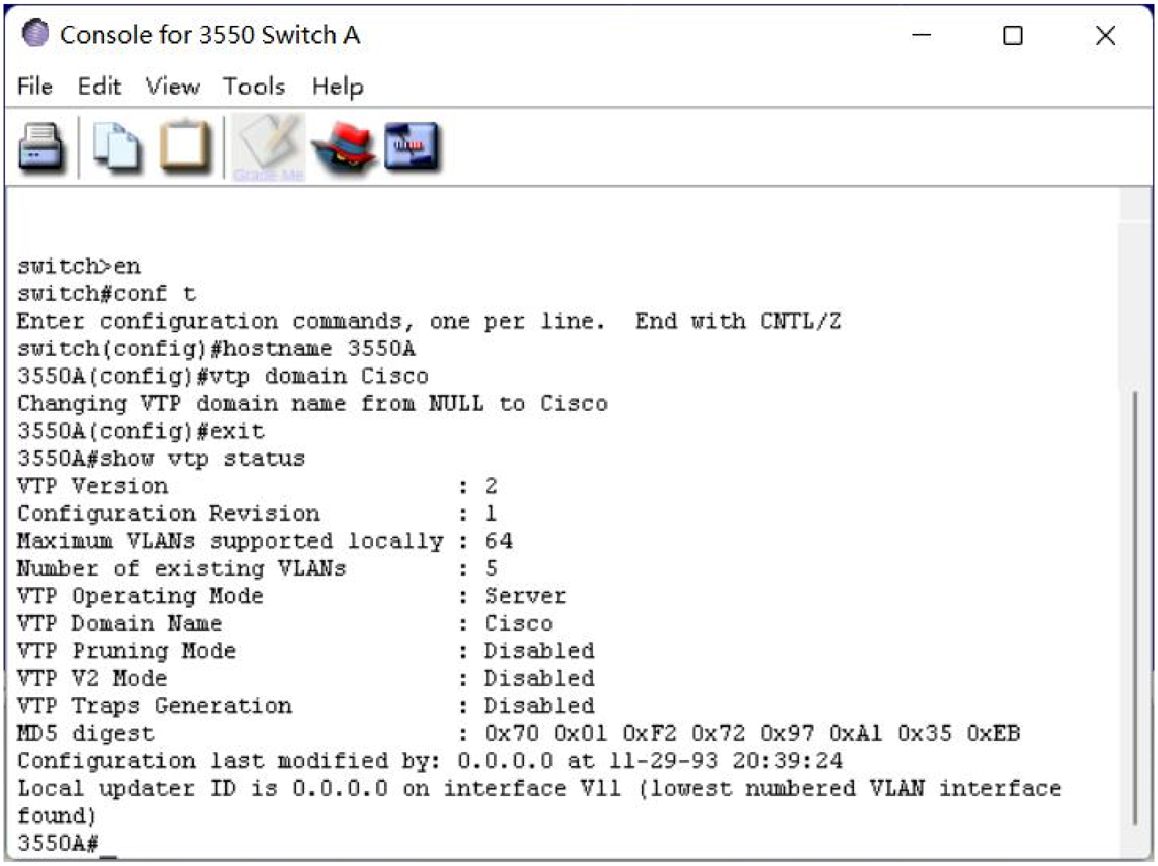
为路由器 A 创建 RIP 协议。



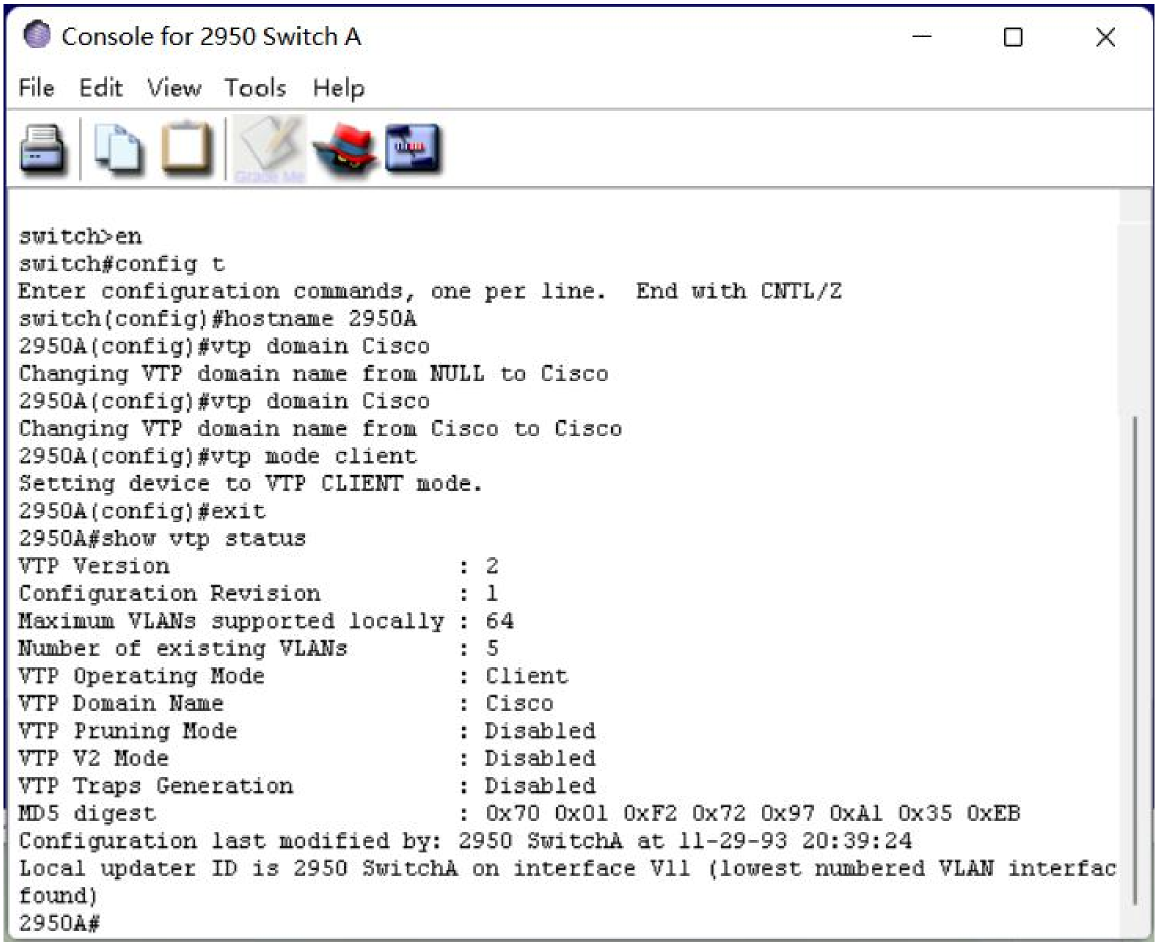
查看路由器 B 路由表，通过 RIP 协议学习到两个新的转发地址。

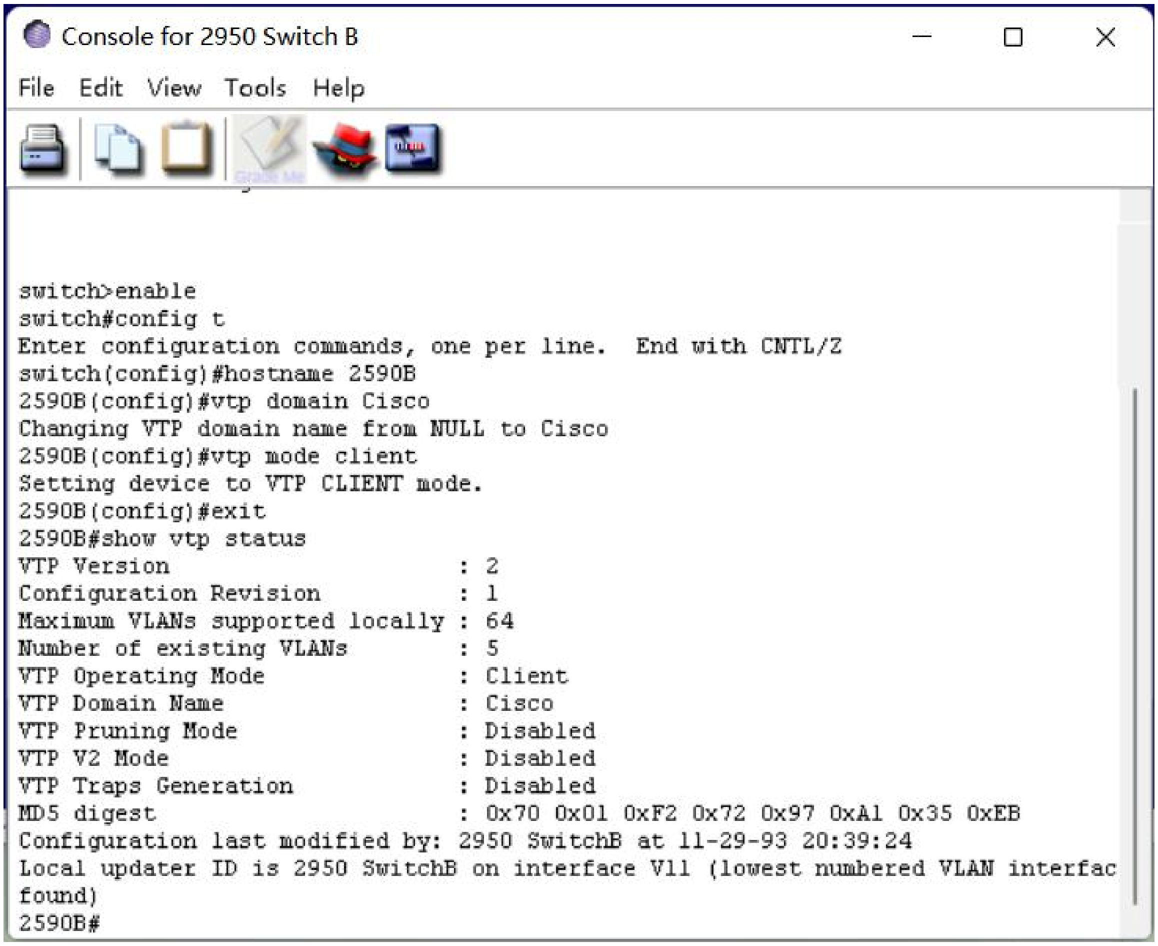


创建新的网络，连接交换机和主机。



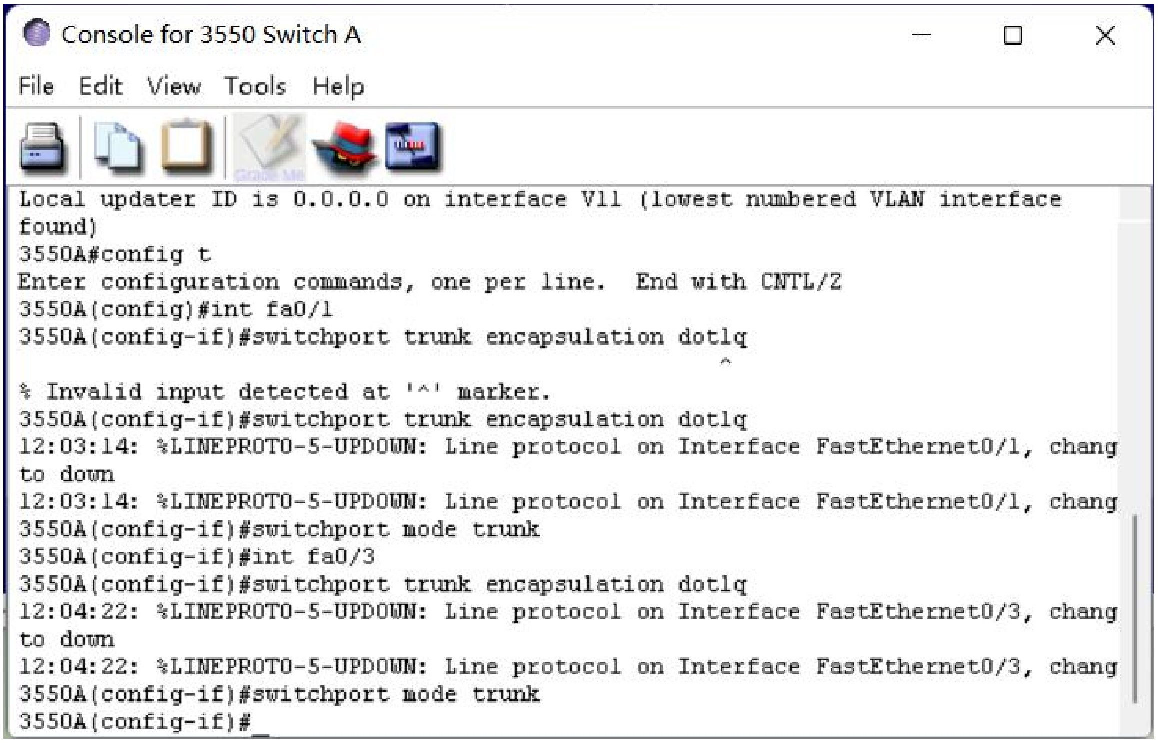
修改 3550 交换机 A 的 VTP 域名称为“Cisco”。


Console for 2950 Switch A
File Edit View Tools Help
switch>en
switch#config t
Enter configuration commands, one per line. End with CNTL/Z
switch(config)#hostname 2950A
2950A(config)#vtp domain Cisco
Changing VTP domain name from NULL to Cisco
2950A(config)#vtp domain Cisco
Changing VTP domain name from Cisco to Cisco
2950A(config)#vtp mode client
Setting device to VTP CLIENT mode.
2950A(config)#exit
2950A#show vtp status
VTP Version : 2
Configuration Revision : 1
Maximum VLANs supported locally : 64
Number of existing VLANs : 5
VTP Operating Mode : Client
VTP Domain Name : Cisco
VTP Pruning Mode : Disabled
VTP V2 Mode : Disabled
VTP Traps Generation : Disabled
MD5 digest : 0x70 0x01 0xF2 0x72 0x97 0xA1 0x35 0xEB
Configuration last modified by: 2950 SwitchA at 11-29-93 20:39:24
Local updater ID is 2950 SwitchA on interface Vl1 (lowest numbered VLAN interfac
found)
2950A#

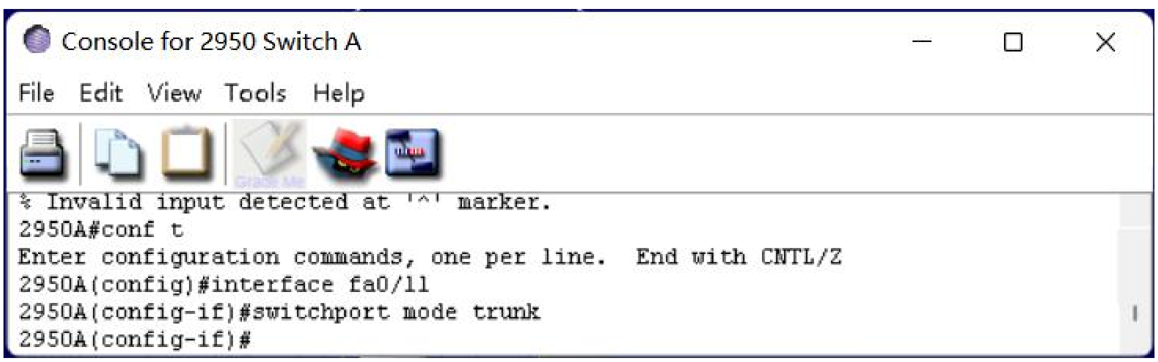


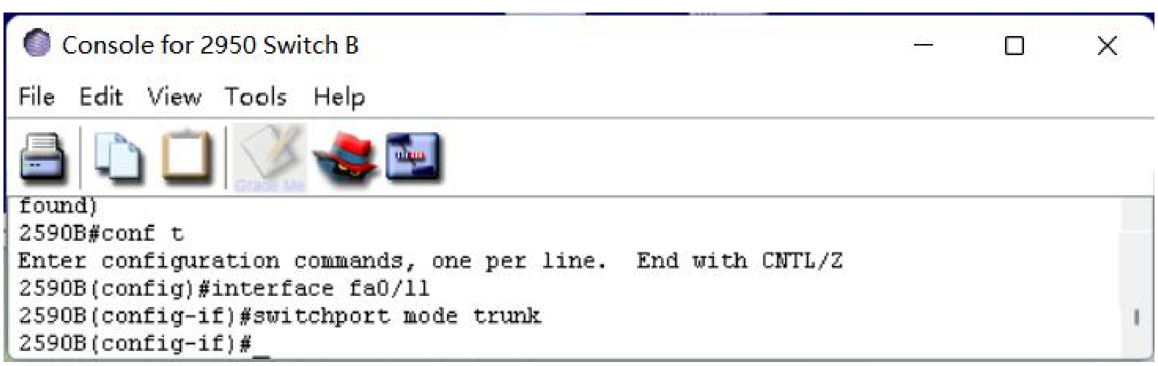

```
switch>enable
switch#config t
Enter configuration commands, one per line.  End with CNTL/Z
switch(config)#hostname 2590B
2590B(config)#vtp domain Cisco
Changing VTP domain name from NULL to Cisco
2590B(config)#vtp mode client
Setting device to VTP CLIENT mode.
2590B(config)#exit
2590B#show vtp status
VTP Version                     : 2
Configuration Revision          : 1
Maximum VLANs supported locally : 64
Number of existing VLANs        : 5
VTP Operating Mode              : Client
VTP Domain Name                 : Cisco
VTP Pruning Mode                : Disabled
VTP V2 Mode                     : Disabled
VTP Traps Generation            : Disabled
MD5 digest                      : 0x70 0x01 0xF2 0x72 0x97 0xA1 0x35 0xEB
Configuration last modified by: 2950 SwitchB at 11-29-93 20:39:24
Local updater ID is 2950 SwitchB on interface Vl1 (lowest numbered VLAN interfac
found)
2590B#
```

同样修改 2950 交换机 A 和 B，将模式修改为客户模式。

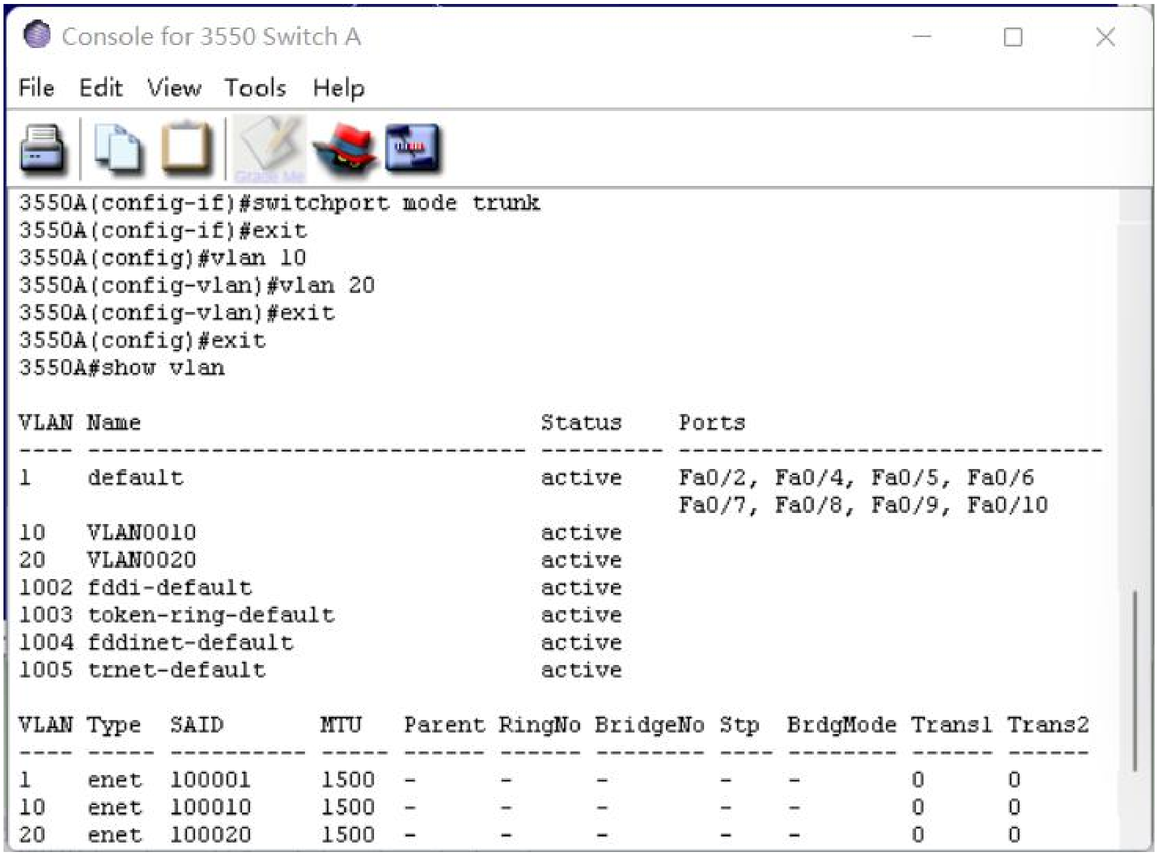


将 3550 交换机 A 和其他两个交换机相连的端口配置为 Trunk 端口。



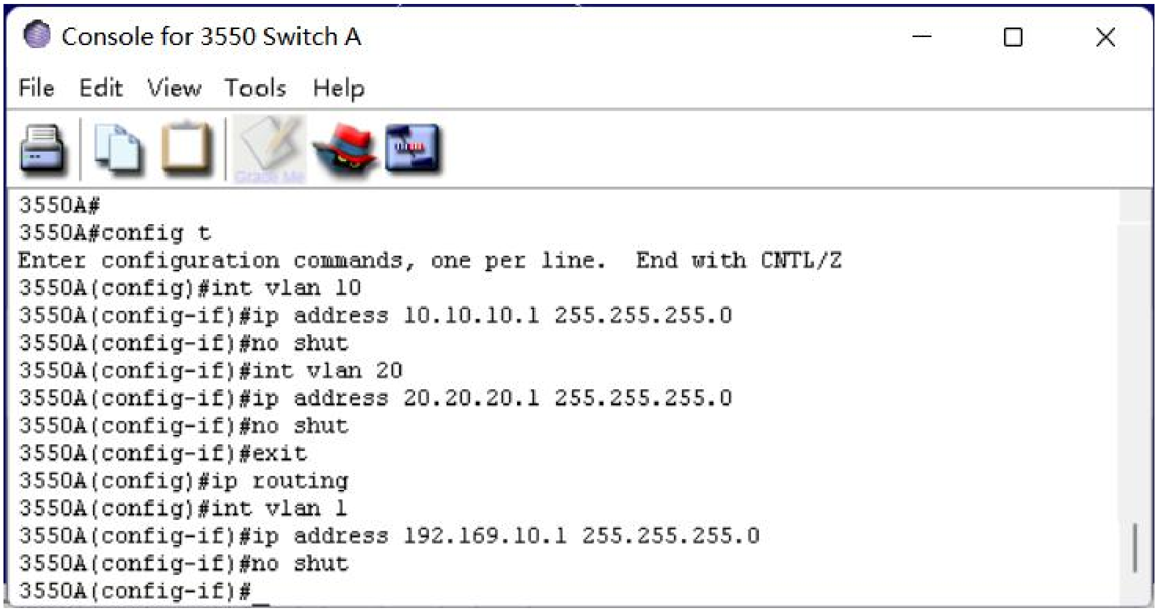


将 2950 交换机 A 和 B 与 3550 交换机 A 相连的端口配置为 Trunk 端口。

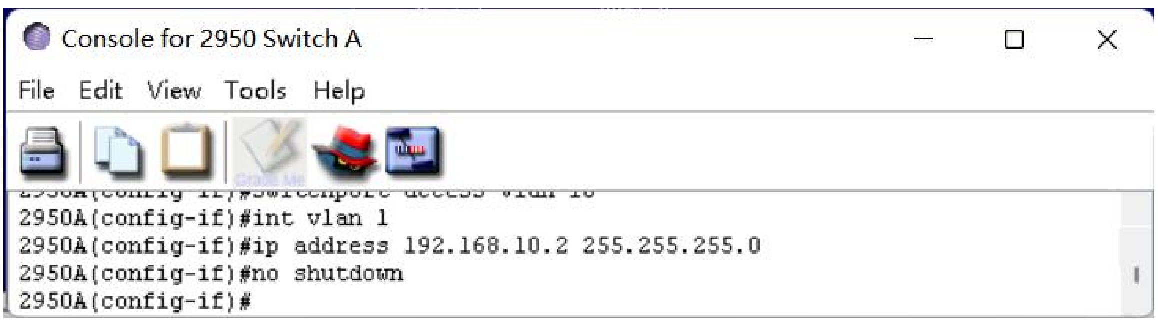


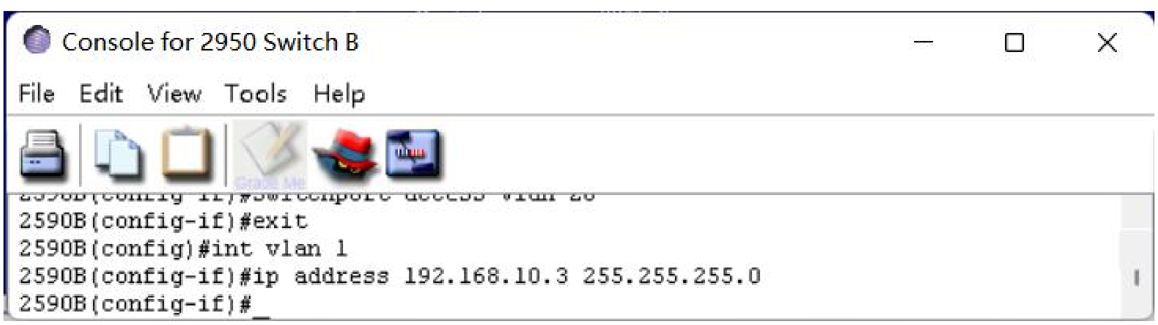
在 3550 交换机 A 上配置 VLAN。

将 2950 交换机 A 和 B 所连设备加入 VLAN。

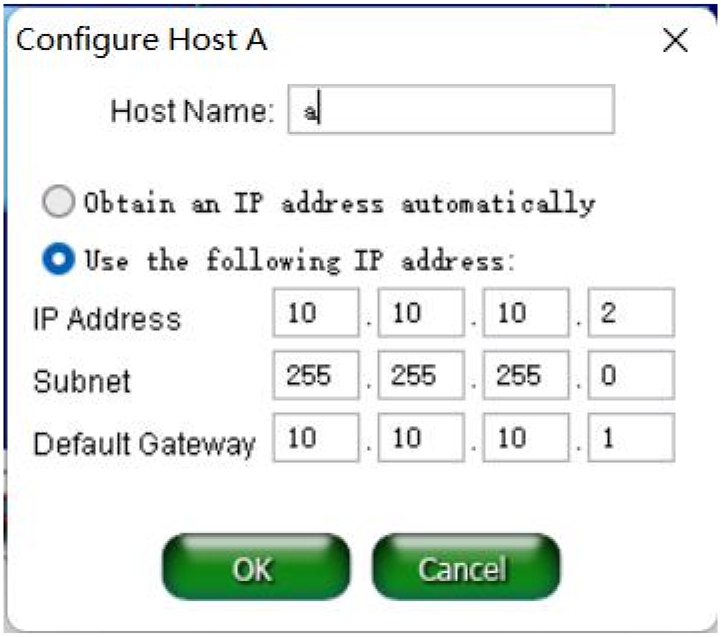


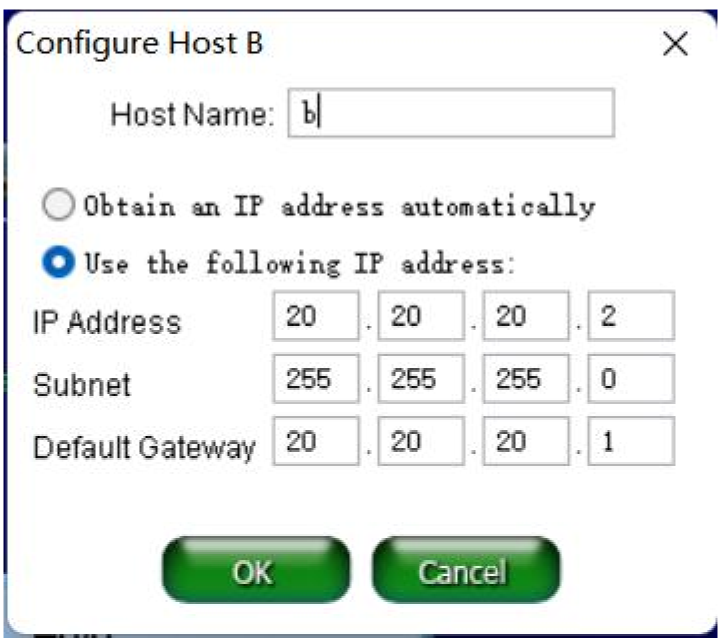
给 3550 交换机 A 的 VLAN 端口配置 IP 地址。



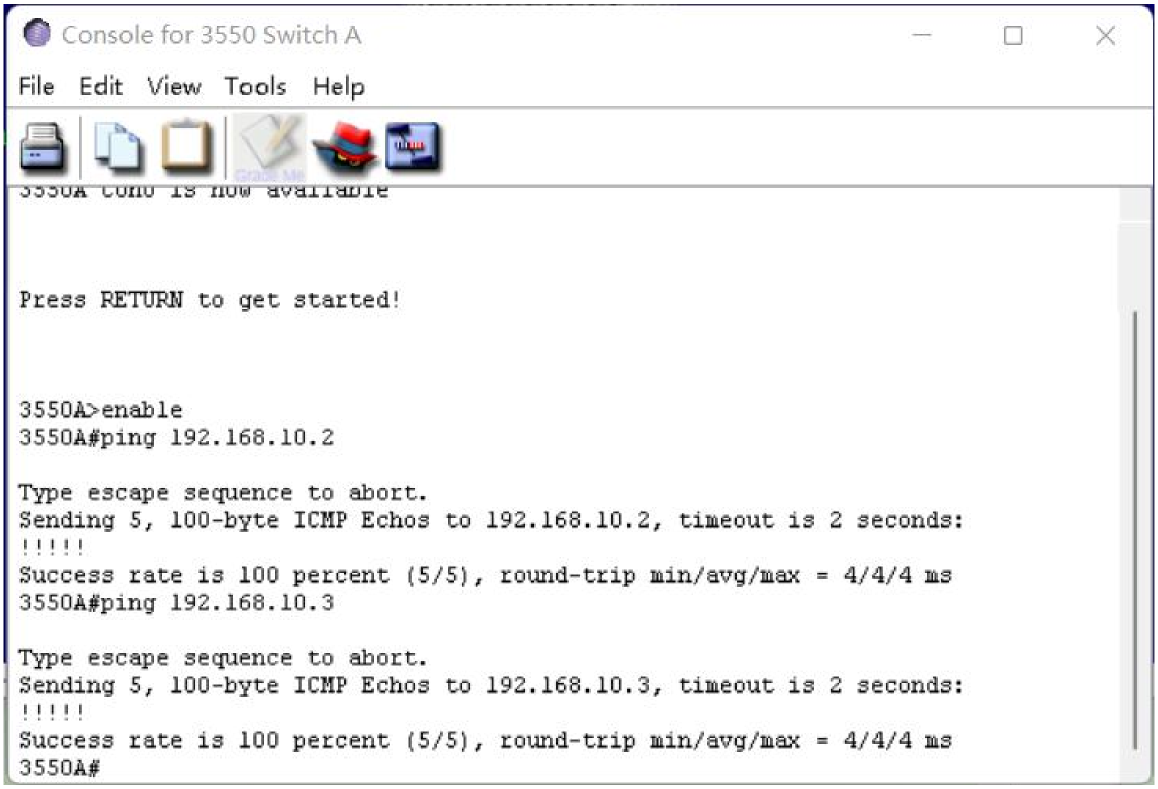


在 2950 交换机 A 和 B 中配置 IP 地址。

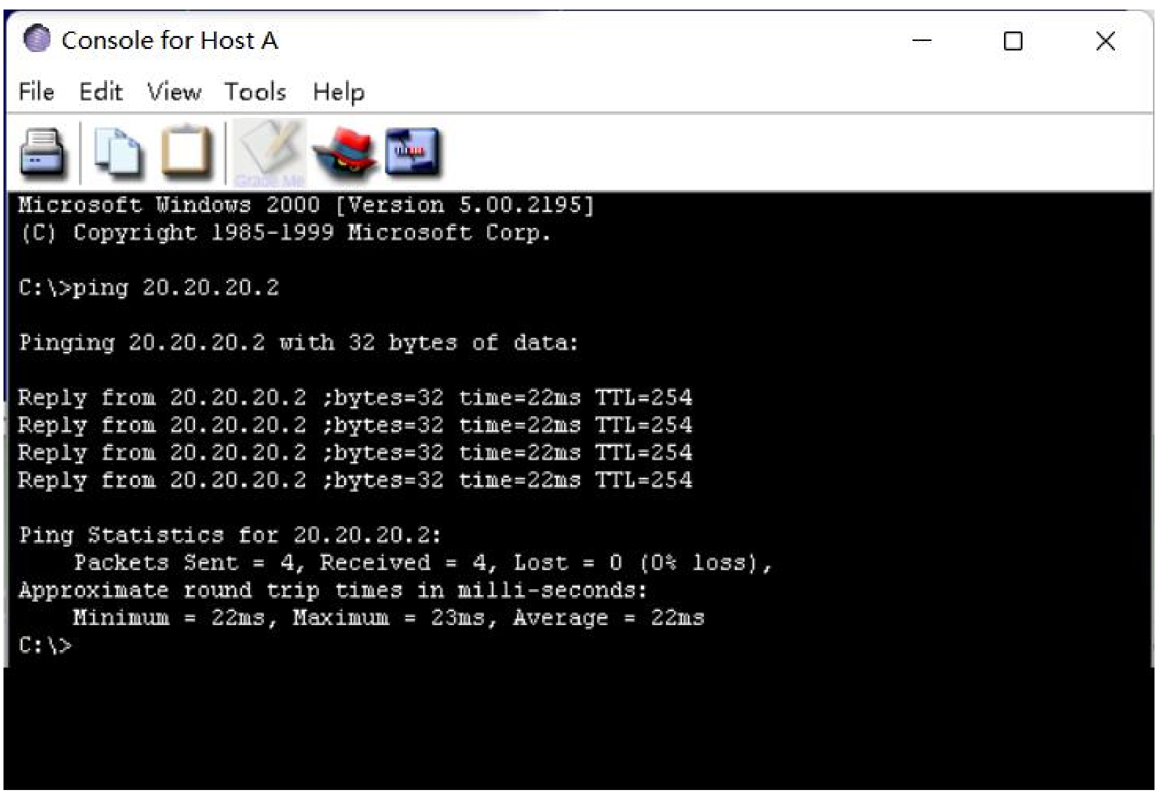




为两台主机分配 IP 地址、子网掩码和默认网关。

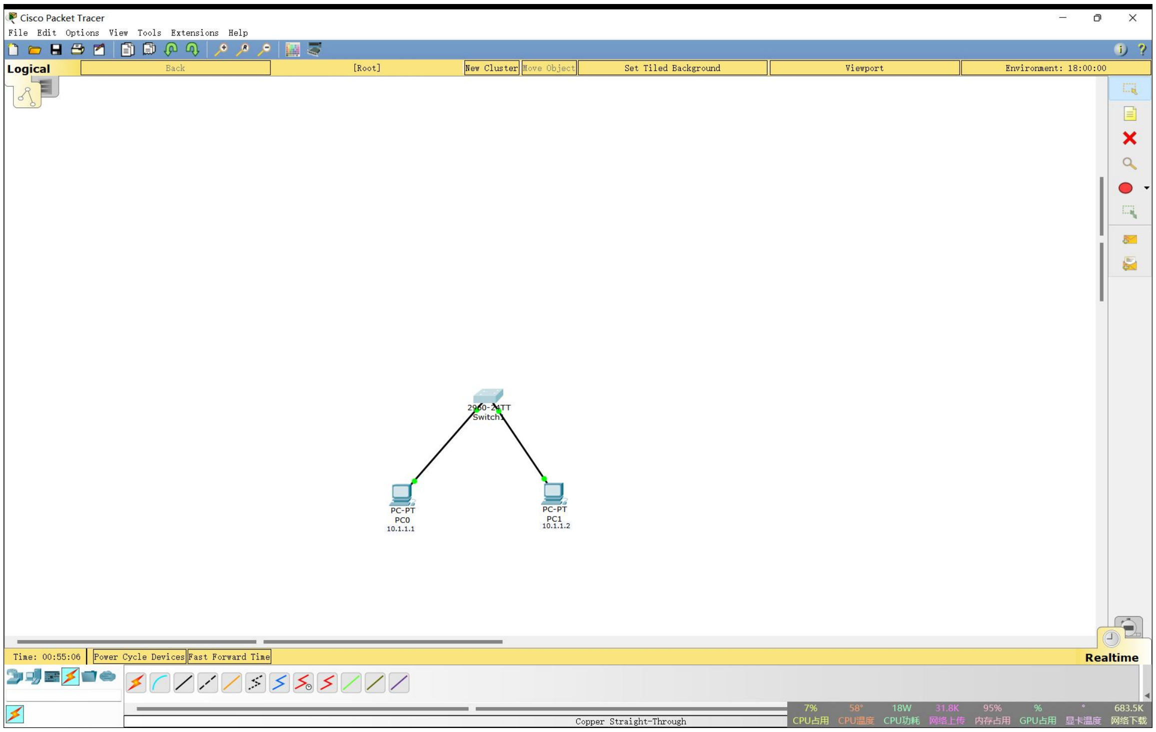


在 3550 交换机 A 上 ping2950 交换机 A 和 B。

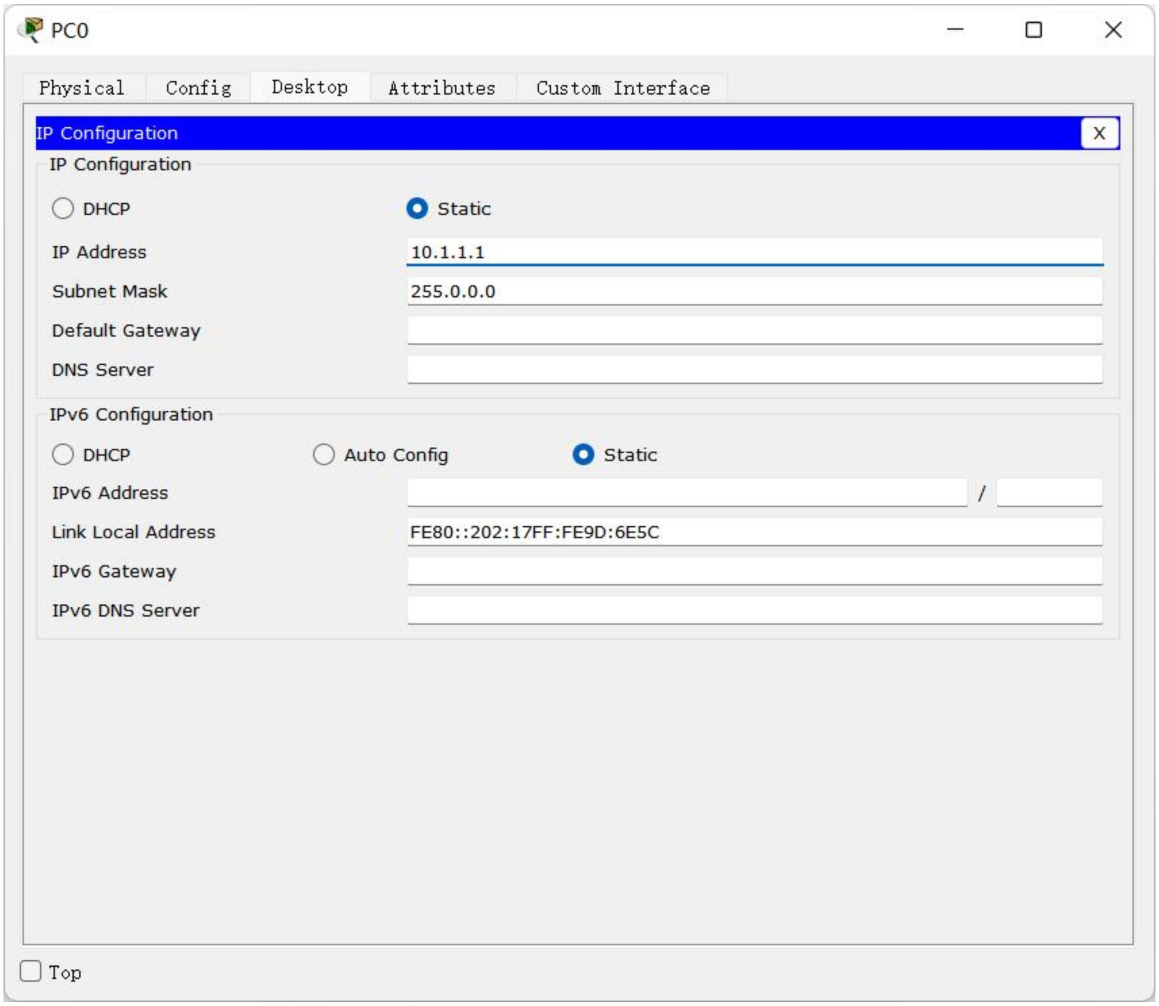


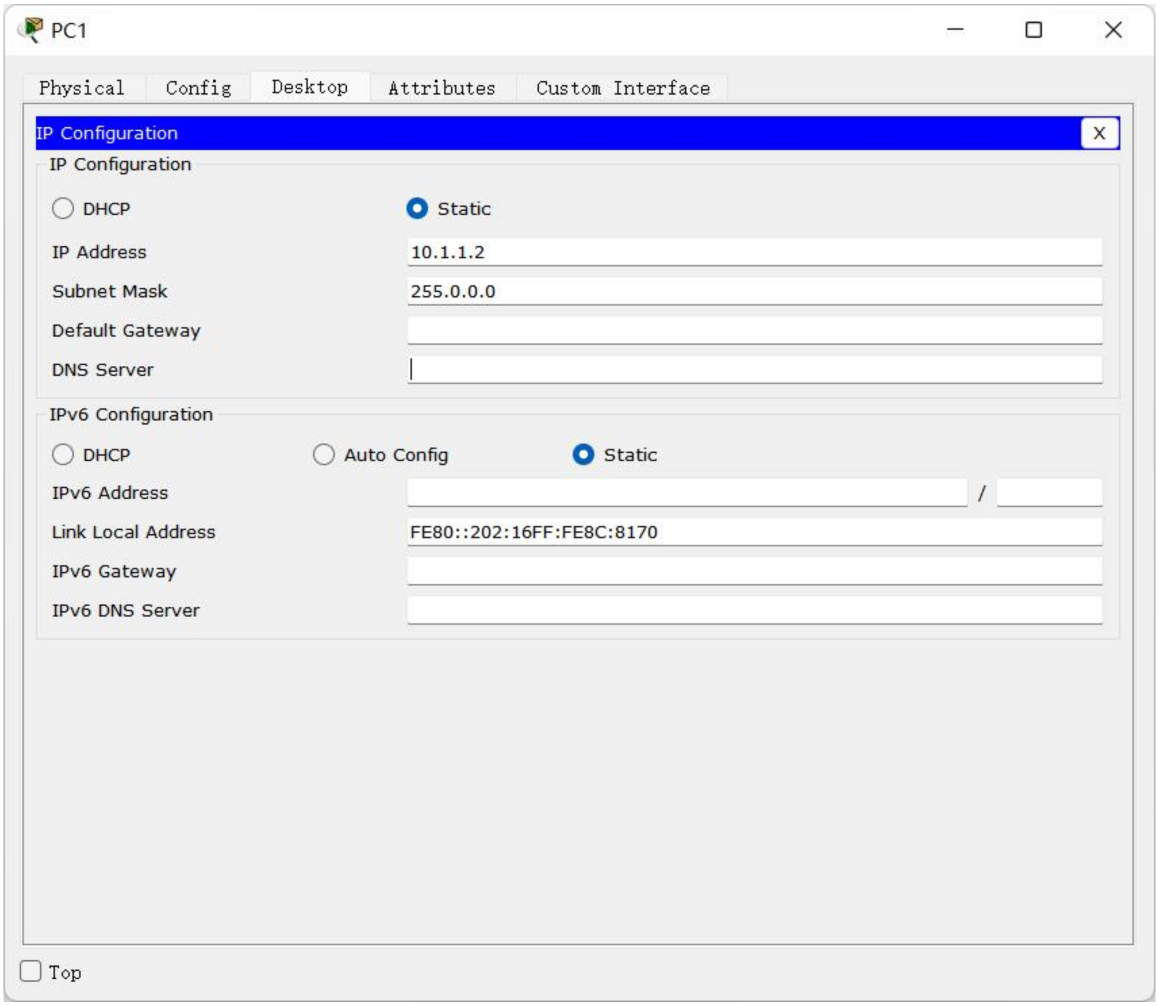
在主机 A 上 ping 主机 B。

3. 按照附件二视频介绍思科模拟器 Packet Tracer 7.0 使用，配置静态路由，配置各种网络设备组网的综合实验。



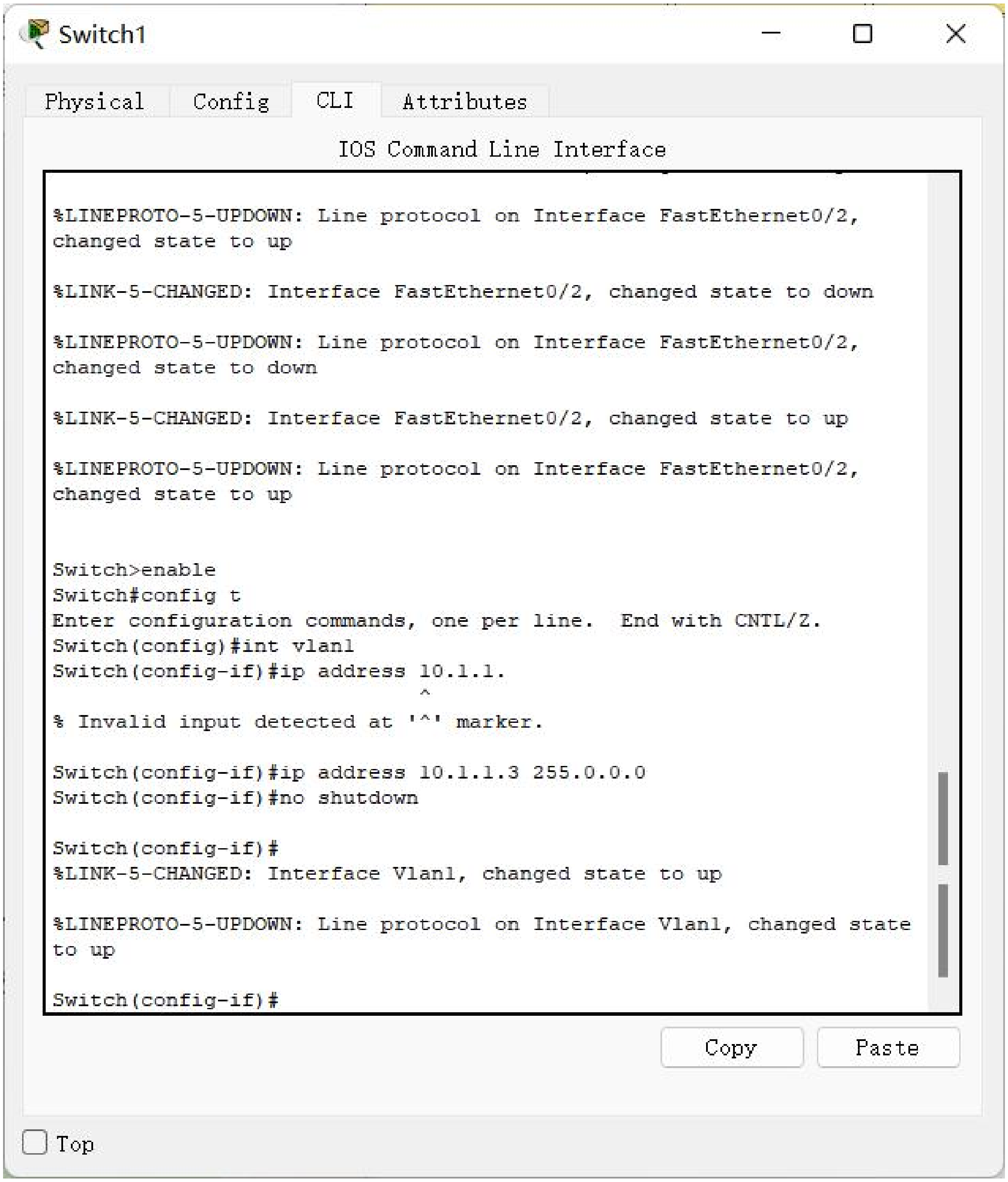
连接交换机和主机。


PC0
Physical
Config
Desktop
Attributes
Custom Interface
IP Configuration
IP Configuration
DHCP
Static
IP Address
10.1.1.1
Subnet Mask
255.0.0.0
Default Gateway
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
Link Local Address
FE80::202:17FF:FE9D:6E5C
IPv6 Gateway
IPv6 DNS Server
Top

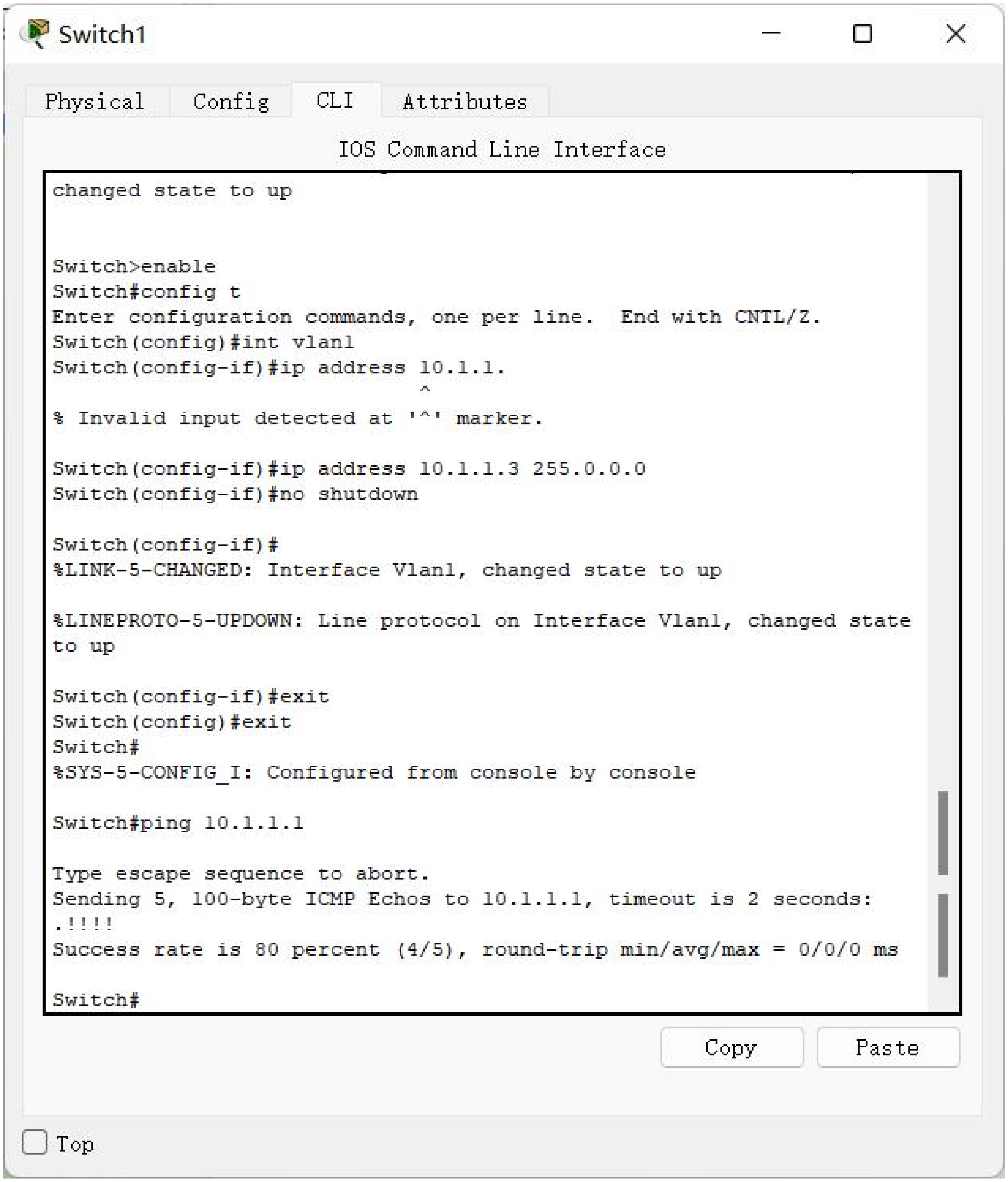


PC1
Physical
Config
Desktop
Attributes
Custom Interface
IP Configuration
X
IP Configuration
DHCP
Static
IP Address
10.1.1.2
Subnet Mask
255.0.0.0
Default Gateway
DNS Server
IPv6 Configuration
DHCP
Auto Config
Static
IPv6 Address
/
Link Local Address
FE80::202:16FF:FE8C:8170
IPv6 Gateway
IPv6 DNS Server
Top

为主机配置 ip 地址。

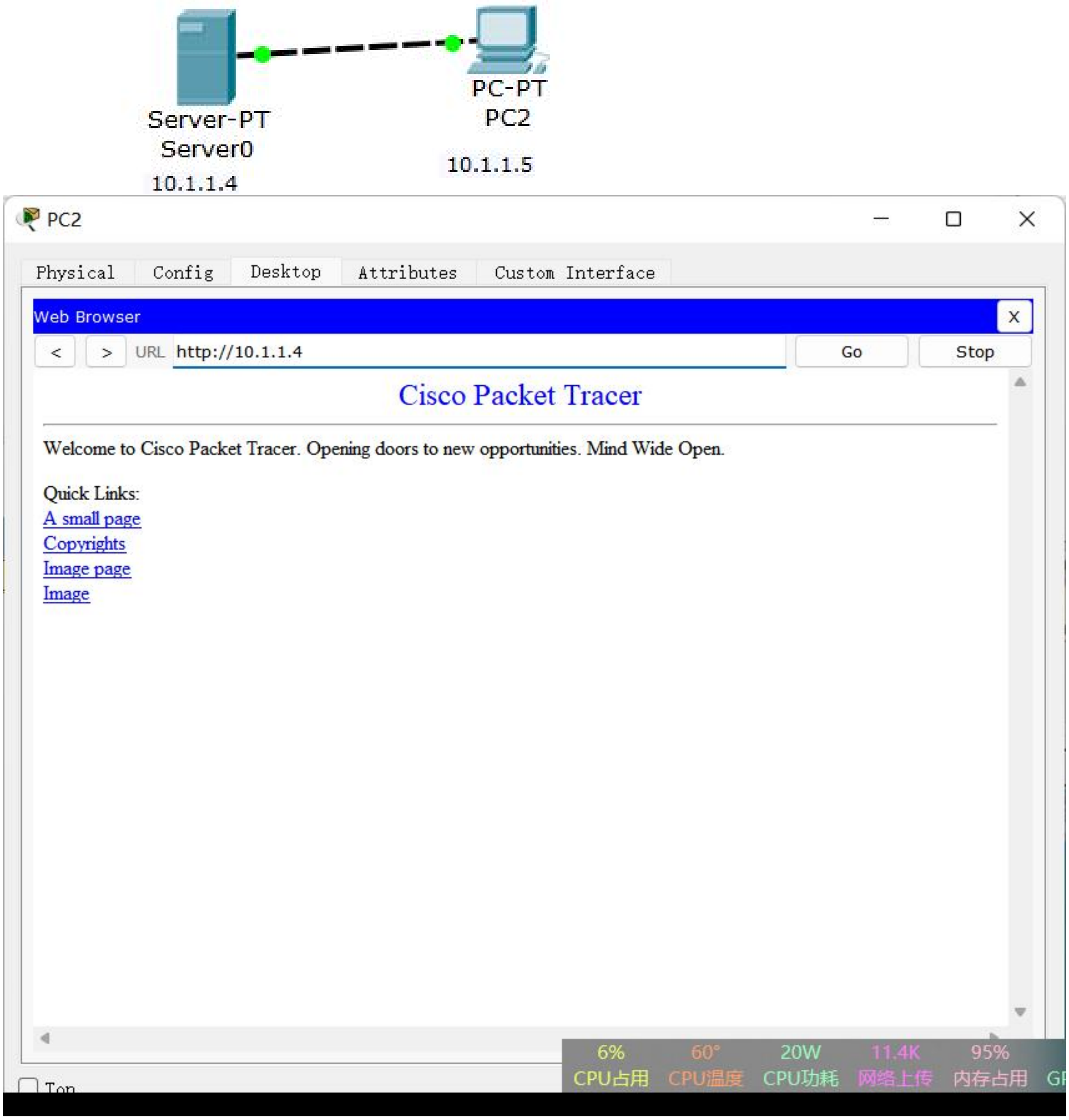

Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int vlan1
Switch(config-if)#ip address 10.1.1.
^
% Invalid input detected at '^' marker.
Switch(config-if)#ip address 10.1.1.3 255.0.0.0
Switch(config-if)#no shutdown
Switch(config-if)#
%LINK-5-CHANGED: Interface Vlan1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan1, changed state to up
Switch(config-if)#
Copy
Paste
Top

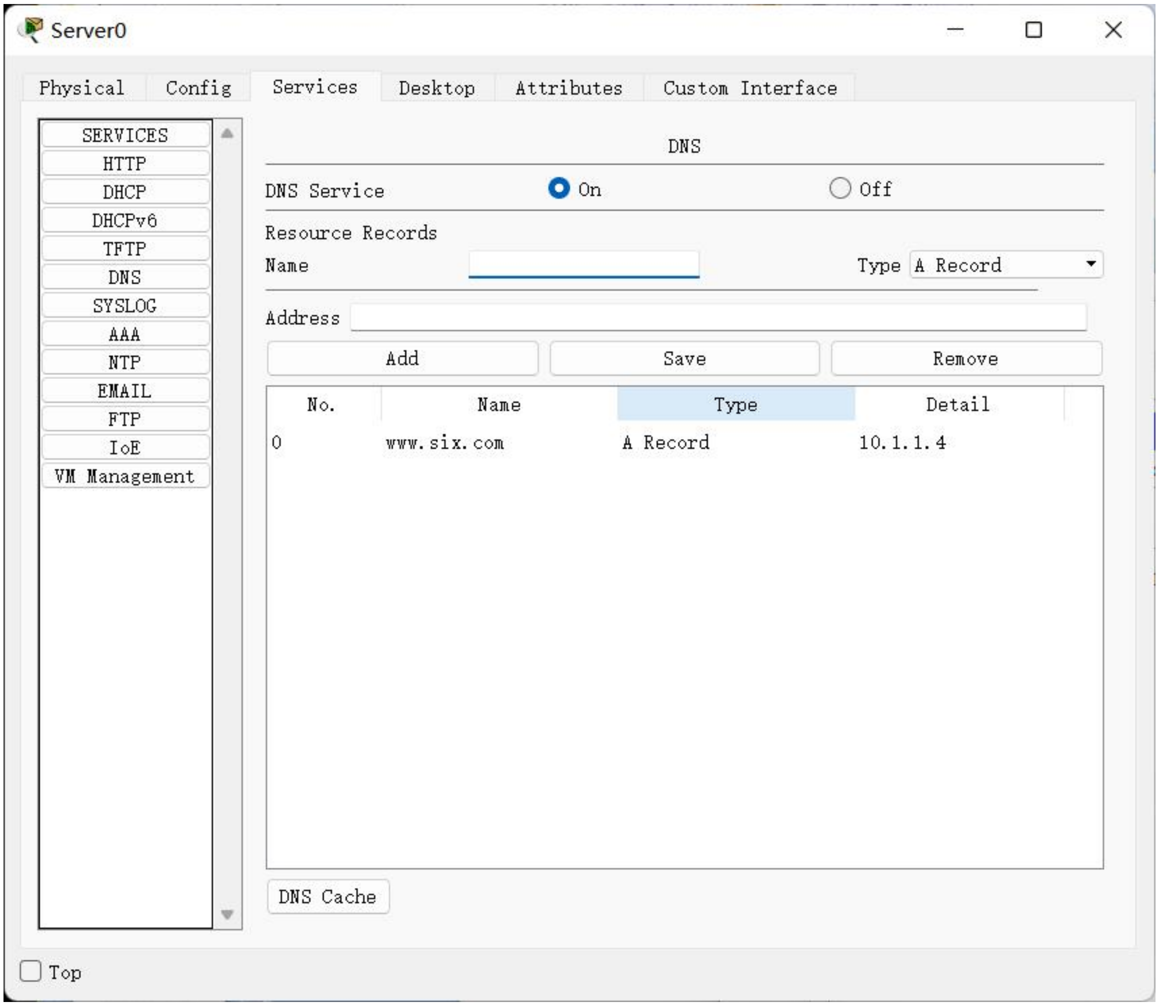
为交换机配置 ip 地址。

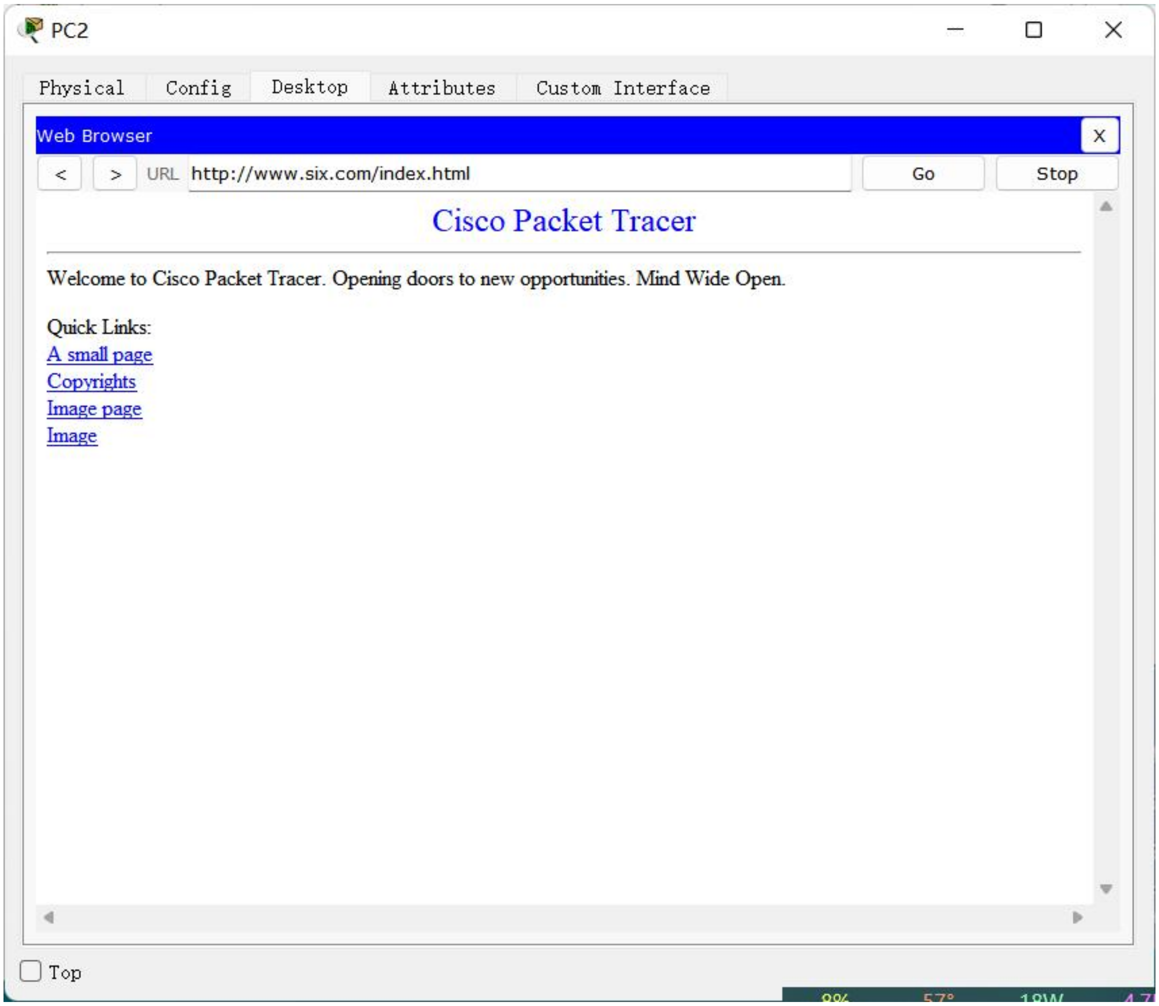


使用 ping 命令查看交换机与主机的连接，第一条数据报被用于 ARP 协议所以没有发送成功。

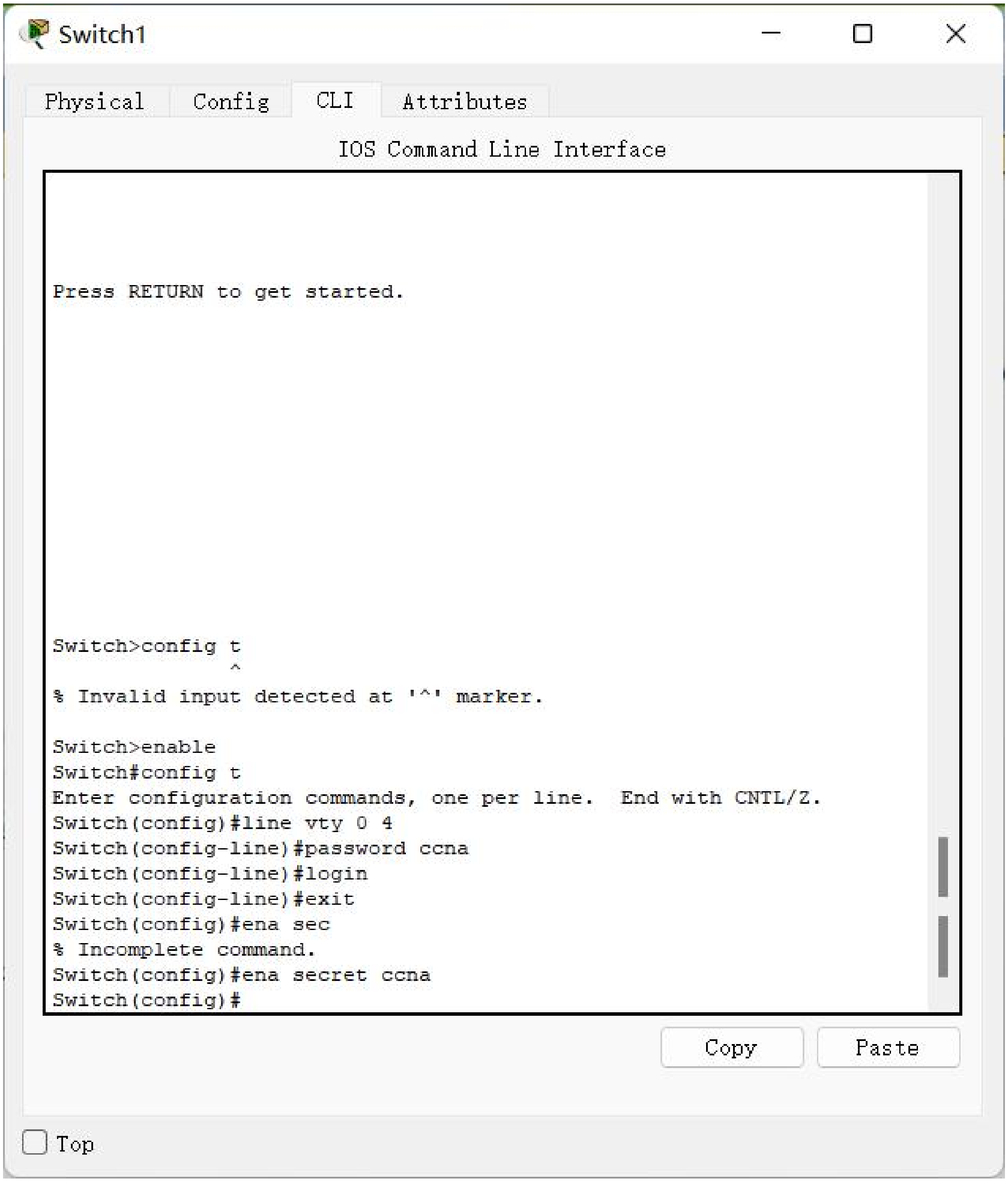
以下是附件二视频中的第一个实验截图。


Server-PT
Server0
10.1.1.4
PC-PT
PC2
10.1.1.5
PC2
Physical
Config
Desktop
Attributes
Custom Interface
Web Browser
URL http://10.1.1.4
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image

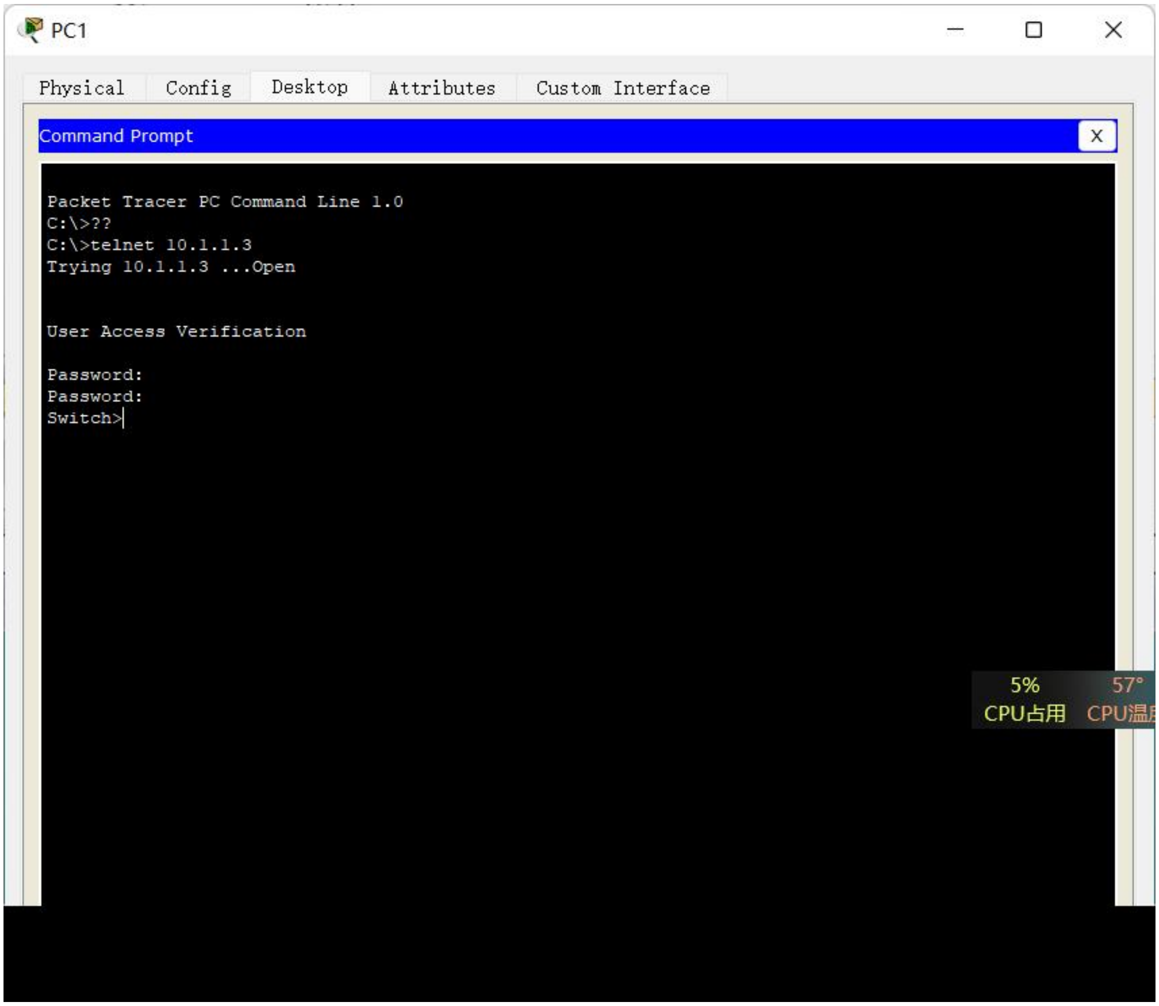

Server0
Physical
Config
Services
Desktop
Attributes
Custom Interface
SERVICES
HTTP
DHCP
DHCPv6
TFTP
DNS
SYSLOG
AAA
NTP
EMAIL
FTP
IoE
VM Management
DNS
DNS Service
On
Off
Resource Records
Name
Type
A Record
Address
Add
Save
Remove
No.
Name
Type
Detail
0
www.six.com
A Record
10.1.1.4
DNS Cache
Top


PC2
Physical
Config
Desktop
Attributes
Custom Interface
Web Browser
URL http://www.six.com/index.html
Go
Stop
Cisco Packet Tracer
Welcome to Cisco Packet Tracer. Opening doors to new opportunities. Mind Wide Open.
Quick Links:
A small page
Copyrights
Image page
Image
Top

设置密码。


Switch1
Physical
Config
CLI
Attributes
IOS Command Line Interface
Press RETURN to get started.
Switch>config t
^
% Invalid input detected at '^' marker.
Switch>enable
Switch#config t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#line vty 0 4
Switch(config-line)#password ccna
Switch(config-line)#login
Switch(config-line)#exit
Switch(config)#ena sec
% Incomplete command.
Switch(config)#ena secret ccna
Switch(config)#
Copy
Paste
Top

连接交换机。



4 实验代码

本次实验的代码已上传于以下代码仓库：https://gitee.com/koshistar/cni-exp/tree/master/E4_3780。

5 课后思考题

无

6 实验总结

本次实验初次使用了模拟软件，加深了对网络层和链路层的理解。

思科模拟器功能丰富且强大，相关实验内容过多便不再进行演示。